

Teacher Notes — Introduction to Functions

Worksheet II — DO NOT DISTRIBUTE — 3 Hours

What students already know coming in

- Set theory: sets, membership, Cartesian product $A \times B$, subsets, set operations, cardinality $|A|$.
- Cantor sheet: bijections informally, countable vs. uncountable (may not be fully internalized yet).
- The maximal interpretation habit — use this when introducing injective/surjective.
- $f(x)$ notation is **brand new**. Do not assume any prior exposure.

Hour 1 — What Is a Function? (~60 min)

Concept block answers

ANSWER

Function (Definition 1.1.6, Bartle & Sherbert): Let A and B be sets. A **function** from A to B is a set f of ordered pairs in $A \times B$ such that for each $a \in A$ there exists a *unique* $b \in B$ with $(a, b) \in f$.

Note for class: this is the set-theoretic definition. The informal version — “a rule that assigns to each element of A exactly one element of B ” — is intuitive but imprecise. B&S use the ordered-pairs form to make the definition unambiguous. This is also why Cartesian products came first.

- Written $f : A \rightarrow B$ (read “ f maps A to B ”).

Intuition: Every input gets exactly one output. No input is left out; no input gets two answers.

Notation blank: $f : A \rightarrow B$

TEACHER NOTE — Introduce with the Minecraft bridge

Start with the Minecraft bedroom example the students already know. A function is a rule that takes a room name and gives back exactly one property.

“What if a room could have two different numbers of beds? Then the rule breaks — it isn’t a function anymore.”

This directly mirrors: one input $\rightarrow$ two outputs = NOT a function.

Four representations — what to put in each box

ANSWER

Arrow diagram: Draw two ovals; draw one arrow from each element of A to exactly one element of B .

Ordered pairs: List as a set $\{(1, a), (2, b), (3, a)\}$ — connects directly to Cartesian product from Set Theory.

Table: Input column | output column. Each input row has exactly one output.

Graph: Plot $(x, f(x))$ pairs on coordinate axes.

TEACHER NOTE — Cartesian product connection — say this explicitly

A function $f : A \rightarrow B$ is a *subset* of $A \times B$.

Special condition: every element of A appears as a first element *exactly once*.

Non-function in ordered-pair form: $\{(1, a), (1, b), (2, c)\}$ — the element 1 appears twice (two arrows from one input).

“This is why we built Cartesian products last week. They’ll feel the payoff here.”

What breaks the definition

ANSWER

The only thing that breaks it: one input mapping to **two or more outputs**.

- One input with no output is also invalid, but focus on the two-output case first — it’s more intuitive.
- Two *different* inputs mapping to the *same* output is fine. (This is the injective distinction — preview it briefly.)

Vertical Line Test answers

ANSWER

Definition: A graph represents a function if and only if every vertical line intersects it *exactly once*.

Note: B&S use “exactly once,” not “at most once.” Since every $a \in A$ must produce an output, a vertical line at any $x = a$ in the domain must hit the graph — so “at most once” would allow missing inputs, which violates the definition. “Exactly once” is the correct form.

Intuition: Each x -value (input) has exactly one y -value (output). A vertical line at $x = 3$ hitting the graph twice means $x = 3$ gives two outputs — not a function.

Passing examples: $y = mx + b$; $y = x^2$ (parabola opening up or down).

Failing example: Circle $x^2 + y^2 = 1$ — at $x = 0$ both $y = 1$ and $y = -1$ lie on the curve.

Set theory fill-in answers

ANSWER

A function $f : A \rightarrow B$ is a subset of $A \times B$.

Special because: each first element appears **exactly once**.

TEACHER NOTE — Timing check — end of Hour 1

By the 60-min mark students should have: the definition, all four representations drawn in their notes, 2–3 non-examples discussed, and the Cartesian product connection filled in.

If running long: cut to just arrow diagrams and ordered pairs; skip tables and graphs (they appear in Hour 2 practice).

Hour 2 — Notation, Domain, Codomain, Range (~60 min)

$f(x)$ notation — concept block answers

ANSWER

Definition: $f(x)$ is the output of function f when the input is x . Read as “ f of x ”.

- f is the **name** of the function (label on a machine).
- x is the **input** (goes in).
- $f(x)$ is the **output** (comes out).

Intuition: $f(x) = 2x + 1$ means “take x , double it, add 1.” The name f just tells us which machine we’re using.

Evaluation practice answers $f(x) = 2x + 1$

ANSWER

$$f(0) = 2(0) + 1 = 1 \qquad f(3) = 2(3) + 1 = 7 \qquad f(-2) = 2(-2) + 1 = -3 \qquad f(a) = 2a + 1$$

Key point on $f(a)$: x is just a placeholder; any variable works.

TEACHER NOTE — Misconception alert — $f(x)$ vs. multiplication

Students sometimes read $f(x)$ as $f \times x$. Address this early.

Say: “ $f(x)$ is not multiplication. The parentheses here mean *input*, not multiply.”

Analogy: think of f as a machine’s name. $f(3)$ means “run the machine named f on input 3.”

Domain / Codomain / Range — answers

ANSWER

Domain (B&S): The set A of first elements of f , denoted $D(f) = A$. Every element of the domain must appear as a first element in f — no input is allowed to be missing.

Codomain (B&S): The set B in $f : A \rightarrow B$. B&S do *not* give the codomain a special name — they simply call it B and note separately that $R(f) \subseteq B$. The codomain is *declared* when you write $f : A \rightarrow B$; it is not determined by the formula alone.

Range / Image (B&S): The set of all second elements in f : $R(f) := \{f(a) : a \in A\}$. The range is always a subset of the codomain, $R(f) \subseteq B$, but need not equal it.

Key: $R(f) \subseteq B$. They are equal precisely when f is surjective (preview this for Hour 3).

TEACHER NOTE — Set Theory payoff moment — make this explicit

Stop and say: “Look what we just wrote: $\text{Range} \subseteq \text{Codomain}$. That’s a subset relationship — exactly what we studied last week.”

This is the moment the two worksheets connect. the students should feel the vocabulary clicking into place.

If they fill in $\text{Range} \subseteq \text{Codomain}$ confidently, the set theory learning transferred.

Practice table answers**ANSWER**

$$f : \{1, 2, 3\} \rightarrow \{A, B, C, D\}, \quad f(1) = A, \quad f(2) = A, \quad f(3) = B$$

$$\text{Domain} = \{1, 2, 3\} \quad \text{Codomain} = \{A, B, C, D\} \quad \text{Range} = \{A, B\}$$

Note: C and D are in the codomain but not the range — fine.

$$g : \mathbb{R} \rightarrow \mathbb{R}, \quad g(x) = x^2$$

$$\text{Domain} = \mathbb{R} \quad \text{Codomain} = \mathbb{R} \quad \text{Range} = [0, \infty)$$

Note: negative numbers are in the codomain but not the range.

$$h : \{\text{cats}, \text{dogs}\} \rightarrow \{0, 1, 2\}, \quad h(\text{cats}) = 2, \quad h(\text{dogs}) = 2$$

$$\text{Domain} = \{\text{cats}, \text{dogs}\} \quad \text{Codomain} = \{0, 1, 2\} \quad \text{Range} = \{2\}$$

Note: 0 and 1 are in the codomain but not the range. Range is a singleton!

TEACHER NOTE — Timing check — end of Hour 2

By the 2-hour mark: $f(x)$ notation, evaluation, domain/codomain/range distinctions, and the subset relationship should all be in students’ notes.

The practice table is the real test. If h above confuses them (range = $\{2\}$ only), spend extra time before moving on.

Hour 3 — Injective, Surjective, Bijective (~60 min)

Concept block answers

ANSWER

Injective (one-to-one) — B&S Definition 1.1.9a:

Primary form: f is injective if whenever $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. In other words, distinct inputs always produce distinct outputs.

Proof form (contrapositive): $f(a) = f(b) \Rightarrow a = b$. This is logically equivalent to the primary form, but is what we use in proofs because it is easier to work with equalities than inequalities. To prove injectivity: assume $f(x_1) = f(x_2)$ and show $x_1 = x_2$.

Intuition: no two arrows point to the same target. Every element of B has *at most* one arrow hitting it.

Diagram: $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$. Arrows: $1 \rightarrow a$, $2 \rightarrow b$, $3 \rightarrow c$. Each target hit at most once. d is unhit — fine.

Surjective (onto) — B&S Definition 1.1.9b:

Definition: f is surjective if $f(A) = B$ (the range equals the codomain). Equivalently: for every $b \in B$, there exists $a \in A$ such that $f(a) = b$.

Intuition: no element of B is left without an arrow. Every target is covered.

Diagram: $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$. Arrows: $1 \rightarrow a$, $2 \rightarrow b$, $3 \rightarrow c$, $4 \rightarrow a$. Every element of B hit.

Bijective:

Definition: both injective and surjective.

Intuition: a perfect one-to-one pairing. Every input has a unique output AND every output is covered.

Requires $|A| = |B|$. Diagram: $A = \{1, 2, 3\}$, $B = \{a, b, c\}$. Arrows: $1 \rightarrow a$, $2 \rightarrow b$, $3 \rightarrow c$. Perfect pairing.

TEACHER NOTE — Draw arrow diagrams BEFORE introducing the vocabulary

Draw these three diagrams on the board first, then ask “What’s different about each one?” Then introduce the formal names.

Diagram 1: $A = \{1, 2\} \rightarrow B = \{a, b, c\}$. Arrows: $1 \rightarrow a$, $2 \rightarrow b$. Element c unhit.

Diagram 2: $A = \{1, 2, 3\} \rightarrow B = \{a, b\}$. Arrows: $1 \rightarrow a$, $2 \rightarrow a$, $3 \rightarrow b$.

Diagram 3: $A = \{1, 2, 3\} \rightarrow B = \{a, b, c\}$. Arrows: $1 \rightarrow a$, $2 \rightarrow b$, $3 \rightarrow c$.

This is the maximal interpretation habit applied to math: look at all the structure before naming it.

Quick-check diagram answers

ANSWER

Diagram 1 — $A = \{1, 2\} \rightarrow B = \{a, b, c\}$. $1 \rightarrow a$, $2 \rightarrow b$. c unhit. **Label: Injective, not surjective.**

Diagram 2 — $A = \{1, 2, 3\} \rightarrow B = \{a, b\}$. $1 \rightarrow a$, $2 \rightarrow a$, $3 \rightarrow b$. **Label: Surjective, not injective.**

Diagram 3 — $A = \{1, 2, 3\} \rightarrow B = \{a, b, c\}$. $1 \rightarrow a$, $2 \rightarrow b$, $3 \rightarrow c$. **Label: Bijective.**

Cantor connection fill-in answers

ANSWER

A bijection between two sets means they have the same **cardinality** (size).

Cantor used this idea to compare the sizes of **infinite** sets.

- $\mathbb{Z}$ is countable: there exists a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$ (list: $0, 1, -1, 2, -2, \dots$).
- $\mathbb{R}$ is uncountable: no such bijection exists (Cantor's diagonal argument, see historian sheet).

TEACHER NOTE — Optional 5-min Cantor discussion

If time allows, revisit the historian sheet briefly.

“Last week we read that Cantor showed $\mathbb{R}$ is bigger than $\mathbb{N}$. Now we have the language to say what that means: there is no bijection from $\mathbb{N}$ to $\mathbb{R}$.”

Not required for the homework, but rewards students who engaged with the sheet.

Challenge problem answers

ANSWER

C1. $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$. $|A| = 3 < |B| = 4$.

(a) Can $f : A \rightarrow B$ be surjective?

No. Surjective requires every element of B to be hit. $|A| = 3 < |B| = 4$, so at least one element of B will always be left without an arrow.

(b) Can $f : A \rightarrow B$ be injective?

Yes. Example: $f(1) = a$, $f(2) = b$, $f(3) = c$. Each input maps to a different target. (d is unhit — fine for injectivity.)

ANSWER

C2. $f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2.$

(a) Is f injective?

No. Counterexample: $f(2) = 4 = f(-2)$, but $2 \neq -2$. One output (4) is hit by two different inputs.

(b) Is f surjective?

No. $f(x) = x^2 \geq 0$ for all real x . So $-1 \in \text{Codomain}$ but has no preimage. $\text{Range} = [0, \infty) \neq \mathbb{R} = \text{Codomain}.$

TEACHER NOTE — C2 follow-up is the key lesson of Hour 3

After reviewing C2, ask: “What if we changed the codomain to $[0, \infty)$?”

→ Then f would be surjective but still not injective.

Then ask: “What if we also changed the domain to $[0, \infty)$?”

→ Then f would be bijective. Same formula, different sets, different classification.

This shows that injective/surjective/bijective are *relationships* between a function AND its domain/codomain — not properties of the formula alone.

End-of-class debrief (5 min)

Ask each student to close their notes and answer from memory:

1. What is the one thing that makes something *not* a function?
2. What is the difference between codomain and range?
3. Give me one example of a function that is injective but not surjective.

If they can answer all three without looking, the concepts landed. Send C1 and C2 home if not finished; they appear on the homework.